

Summary. This multimodal running example illustrates how epistemic friction arises naturally from referential ambiguity, shared visual context, and multi-agent coordination. FAR, FPP, GRFR, and FTR incorporate the same structured friction signal at different points in the learning pipeline, but all induce the same qualitative behavior: defer commitment when the common ground is underspecified, and intervene early to stabilize long-horizon collaboration.

8 Evaluation: Measuring Epistemic Alignment

Prevailing evaluation paradigms in alignment research are largely inherited from static language modeling and preference optimization. Models are typically assessed by single-turn correctness, preference model agreement, or scalar reward scores, even when they are deployed in interactive settings that require sustained epistemic regulation over time.

As a result, these paradigms systematically under-measure core aspects of epistemic competence: the ability to detect underspecification, to manage uncertainty, to repair contradictions, and to intervene proportionally as an interaction unfolds. In such frameworks, hesitation, clarification, and repair are either invisible to the metrics or treated as deviations from optimal behavior, rather than as central objects of evaluation.

Recent work has begun to expose these limitations by moving toward interactive and trajectory-based evaluation. For example, [Nath et al. \(2025\)](#) evaluate frictional agents in collaborative dialogue tasks using contrastive intervention pairs and measure how interventions affect belief alignment and downstream task success over time. Such settings make visible competencies that are systematically obscured by single-turn benchmarks.

However, existing interactive evaluations remain largely task-specific and lack a general, principled set of dimensions for assessing epistemic intervention quality. They demonstrate the need for richer evaluation, but do not yet provide a unifying framework for comparing intervention behavior across domains, tasks, and learning paradigms.

In this section, we propose a unified evaluation framework grounded directly in the taxonomy of frictional interventions and the epistemic control problem developed earlier. Our goal is to define evaluation dimensions that measure not only what

a model answers, but how it manages uncertainty, risk, and commitment over the course of an interaction.

The preceding sections introduced Frictive Policy Optimization as a framework for learning epistemically responsible intervention policies. We now turn to the complementary question of how such policies should be evaluated. In this section, we address a central methodological question: *How should such policies be evaluated?*

We argue that prevailing evaluation paradigms in alignment research are systematically misaligned with epistemic competence. They emphasize surface-level helpfulness and preference satisfaction, while ignoring the temporal, intervention-based behaviors that FPO is designed to induce. We therefore propose an evaluation framework grounded in *epistemic alignment*: the ability of a model to manage uncertainty, prevent epistemic failure, and regulate commitment over the course of an interaction.

8.1 Evaluation Dimensions from the Intervention Taxonomy

We derive evaluation dimensions directly from the taxonomy of Section 3. Each dimension corresponds to a class of frictional intervention and a characteristic epistemic failure mode. Rather than evaluating surface-level correctness, these metrics target the core control decisions that determine how an agent manages uncertainty, commitment, and risk over time.

This approach follows a growing literature on interactive and epistemic evaluation, which emphasizes clarification, calibration, repair, and proportional refusal as central competencies in dialogue ([Williams and Young, 2007](#); [Young et al., 2013](#); [Gervits et al., 2021](#); [Nath et al., 2025](#)).

Clarification Competence. Clarification competence measures whether a model detects underspecification and asks for missing constraints before acting. This dimension is central to decision-theoretic models of question asking and clarification, where the primary failure mode is premature commitment under uncertainty ([Lindley, 1956](#); [Gervits et al., 2021](#); [Hou et al., 2024](#)).

Let $\mathbf{1}_{\text{nec}}(h)$ indicate that clarification is necessary in context h . Let $\mathbf{1}_{\text{ask}}(h)$ indicate that the model issues a clarificatory intervention. We then define the score associated with providing clarification:

(33) ClarifyScore =

$$\mathbb{E} [\mathbf{1}_{\text{nec}}(h) \mathbf{1}_{\text{ask}}(h)] - \mathbb{E} [(1 - \mathbf{1}_{\text{nec}}(h)) \mathbf{1}_{\text{ask}}(h)]$$

This measures the net rate of appropriate clarification: the first term rewards asking when clarification is necessary, while the second penalizes unnecessary clarification. The score operationalizes sensitivity to underspecification rather than raw intervention frequency.

As a metric, this penalizes both omission (failing to clarify when necessary) and overuse (clarifying when unnecessary), reflecting the classical precision–recall tradeoff in selective querying.

Calibration and Uncertainty Expression. Calibration measures alignment between expressed confidence and empirical correctness. Miscalibration is a well-documented failure mode in modern language models and a key driver of epistemic overcommitment (Niculescu-Mizil and Caruana, 2005; Guo et al., 2017; Jiang et al., 2023).

Let $\text{conf}(y \mid h)$ be a confidence proxy and $\text{acc}(y, h) \in \{0, 1\}$ an indicator of correctness. We define expected calibration error as follows:

(34) ECE =

$$\mathbb{E} [|\text{conf}(y \mid h) - \text{acc}(y, h)|]$$

This measures miscalibration as the expected absolute gap between expressed confidence and empirical correctness. In the present framework, high ECE reflects epistemic overcommitment or unwarranted hedging rather than purely probabilistic error.

Unlike standard calibration metrics, this score is interpreted jointly with intervention behavior. A model may reduce ECE either by hedging or by intervening (clarifying or deferring) rather than committing, making calibration inseparable from intervention policy.

Contradiction Detection and Repair. Contradiction competence measures whether a model detects inconsistencies and repairs them over time. This dimension draws on work in dialogue state tracking and belief revision, where recovery from error is often more important than initial correctness (Traum and Larsson, 2003; Williams and Young, 2007; Young et al., 2013).

Let $\text{Contr}(h_t, y_t)$ be the contradiction score defined in Section 5. Let $\mathbf{1}_{\text{repair}}(t)$ indicate that a repair occurs after time t . We define:

(35) RepairScore =

$$\mathbb{E} \left[\sum_t \mathbf{1}[\text{Contr}(h_t, y_t) > \tau] \cdot \mathbf{1}_{\text{repair}}(t) \right]$$

This measures dynamic epistemic competence: it counts the expected number of times a detected contradiction is followed by an explicit repair action. The metric evaluates recovery from epistemic error rather than static correctness; it identifies not whether errors occur, but whether the agent can restore epistemic coherence after they occur.

Refusal Quality and Proportionality. Refusal is evaluated as a graded intervention, not a binary outcome. Recent work on abstention and selective answering emphasizes that refusal quality, justification, and proportionality are central to epistemic alignment (Wen et al., 2025; Bai et al., 2022; Clark et al., 2025).

Let $\mathbf{1}_{\text{wr}}(h)$ indicate that refusal is warranted in context h . Let $\text{Just}(y)$ score the quality of justification and $\text{Alt}(y)$ the presence of constructive alternatives. We define:

(36) RefusalScore =

$$\mathbb{E} [\mathbf{1}_{\text{wr}}(h) (\text{Just}(y) + \text{Alt}(y))] - \mathbb{E} [(1 - \mathbf{1}_{\text{wr}}(h)) \mathbf{1}_{\text{refuse}}(y)]$$

This metric operationalizes *proportionality of disobedience* rather than raw refusal rate: it discourages blanket refusal while rewarding principled, context-sensitive disobedience. The RefusalScore is normative rather than purely behavioral. Justified, constructive refusals—those that block hazardous, unethical, or epistemically incoherent actions—are treated as productive friction and contribute positively to the net friction functional. In contrast, unnecessary or obstructive refusals that block well-posed, low-risk requests are treated as unproductive friction and are penalized accordingly.

Information Efficiency. Information efficiency measures the epistemic benefit obtained per unit of intervention cost. This dimension is standard in Bayesian experimental design and decision-theoretic question generation, where the goal is to minimize uncertainty with minimal querying (Lindley, 1956; Gervits et al., 2021; Hou et al., 2024).

Let $C_{\text{fric}}(a_t)$ be the intervention cost and $\text{InfoGain}(h_t, a_t)$ the expected entropy reduction. We define:

$$(37) \text{ InfoEff} = \frac{\mathbb{E} [\sum_t \text{InfoGain}(h_t, a_t)]}{\mathbb{E} [\sum_t C_{\text{fric}}(a_t)]}$$

This measures the expected epistemic benefit obtained per unit of interaction cost. High values indicate that the model reduces uncertainty efficiently rather than by excessive or redundant intervention. High scores indicate that the model reduces uncertainty with minimal user burden.

8.2 Benchmark Task Classes and Trajectory Protocol

We propose three classes of benchmark tasks designed to elicit frictive behavior.

Underspecified Instruction Tasks. These tasks omit critical constraints. Success requires issuing a clarificatory intervention before task completion. Evaluation focuses on: clarification necessity detection; minimality of questions; and downstream task success after clarification.

Epistemic Hazard Tasks. These tasks embed latent safety, ethical, or normative hazards, where evaluation measures include: hazard detection accuracy; proportionality of refusal; and the quality of redirection or alternatives.

Multi-Turn Epistemic Repair Tasks. These tasks introduce contradictions or false premises during a multi-party dialogue. Evaluation measures: detection latency; repair correctness; and restoration of epistemic coherence.

For each of these classes, evaluation is conducted over dialogue trajectories rather than single turns. Given a fixed set of initial contexts $\{h_0^{(i)}\}$, each model is rolled out to produce trajectories $\tau^{(i)}$. Metrics are computed per trajectory and averaged across models. Such metrics are informed by formal studies of grounding and belief alignment in multi-agent systems, which emphasize collaborative resolution of uncertainty and shared understanding (Obiso et al., 2025).

We recommend paired evaluation, in which multiple models are exposed to identical contexts, enabling controlled comparison of intervention behavior.

8.3 Revisiting the Running Example

We now revisit the multimodal, multi-agent running example introduced in Section 7, and illustrate how the evaluation dimensions defined above make the epistemic behavior of different FPO methods directly measurable.

In this example, the agent must integrate partial visual evidence, incomplete task constraints, and evolving user intent over multiple turns, while coordinating with another agent toward a shared goal. At several points in the interaction, the agent faces non-trivial intervention choices: whether to answer immediately, request clarification about missing parameters, challenge an unsafe assumption, or defer action until additional evidence is obtained.

A model trained with Frictive Policy Optimization is not evaluated solely on final task success, but on the "trajectory of epistemic control" it exhibits along the way. Concretely, the proposed metrics capture complementary aspects of this behavior:

- *Clarification competence* measures whether the model requests missing visual or task constraints before committing to an action.
- *Calibration and uncertainty expression* evaluate whether the model modulates confidence or intervenes rather than overcommitting under ambiguous perceptual input.
- *Contradiction detection and repair* capture whether inconsistencies across modalities or across turns are explicitly detected and corrected.
- *Refusal quality and proportionality* measure whether the model resists unsafe or ill-posed requests constructively, offering principled justifications and alternatives.
- *Information efficiency* quantifies how much epistemic uncertainty is reduced per unit of interaction cost across the dialogue.

Importantly, these dimensions distinguish between superficially similar behaviors that standard benchmarks conflate. For example, two agents may both achieve the same final task outcome, while differing substantially in how many redundant clarifications they issue, whether they repair early misinterpretations, or whether they intervene proportionally when safety constraints become salient.

In this sense, the running example makes explicit what the proposed evaluation framework is designed to measure: not static response quality,